"MORE THAN WORDS": GENRE-DISCRIMINATING LYRICAL PATTERNS IN ANGLOPHONE GERMAN POPULAR MUSIC

Anonymous Authors
Anonymous Affiliations
anonymous@ismir.net

ABSTRACT

Recent advances in natural language processing have improved lyrics-based music genre classification through large language models, yet such black box systems obscure the linguistic structures that actually differentiate music genres. Earlier research in computational musicology and MIR, meanwhile, has emphasized structural and formal stylistic readings of lyrics without systematically investigating recurrent content-related lyrical, and formulaic expressions in particular—leaving stylistic genre differences remain under-researched until now. Our study contributes to filling this gap by uncovering genre discriminative lyrical patterns—topics, sentiments, and formulaic expressions—in fifty years of Anglophone German popular music. Using the POPTRAG corpus and *Discogs* genre annotations for 126,042 tracks with English lyrics, we apply the fightin' words method to identify statistically distinctive n-grams across genres and build interpretable topic, sentiment, and expression vocabularies. Finally, we identify latent types of n-grams contained in each vocabulary via structural topic modeling using target genres as prevalence priors. Resulting interpretable lyrics features are employed in machine-learning models that reveal how topics, sentiments, and formulaic expressions function as stylistic markers for single popular music genres. Our transparent ML approach retains roughly 80% of the predictive performance of a transfer-learned DistillBERT classifier while offering conceptual insight into how lyrics encode genre identity.

1. INTRODUCTION

Popular music does not resonate with audiences through sound alone. Lyrics shape emotional interpretation, social meaning, and collective memory [1,2], making them a central component of the listening experience and a rich object of study for Computational Musicology (CM) and Music Information Retrieval (MIR) [3]. Because lyrics are text, they enable large-scale analysis across songs and genres, complementing audio-based approaches with an explicit linguistic perspective on popular music. This makes them especially useful for studying genre, since genre organizes songs through recurring stylistic markers and sociocultural conventions [4], many of which are encoded in lyrics [5–9]. To this end, lyrics-based genre recognition can serve as a methodological lens: by predicting genre from textual features, it can uncover which lyrical patterns most strongly differentiate genres and how these patterns vary across the popular music landscape [10, 11].

Recent research at the intersection of MIR and Natural Language Processing (NLP) has leveraged deep learning architectures and word embeddings to model lyrics for genre classification [10, 12–14]. Pretrained, contextual representation models, such as BERT [15] and its derivatives (e.g., [16, 17]) have become widely used in NLP classification tasks, offering strong predictive performance but still limited transparency [18]. Their embeddings compress complex semantic and syntactic information into high-dimensional spaces whose dimensions lack an interpretable meaning. Consequently, while such models excel at prediction, they provide little understanding of how words and linguistic patterns shape genre distinctions. While there exist approaches to trace black box model decisions back to individual humam-readable token through attention visualisation [10,19]; they do not mitigate semantic bias during embedding of tokens [20] and insights remain at the sparse representation level of individual tokens. This interpretability gap limits both scholarly attempts to connect lyrical style with musical culture and applied uses of lyrics-based classifiers, where understanding model behavior is essential for debugging, bias detection, and trustworthy deployment [21, 22].

More interpretable approaches to lyrics-based music genre analysis in CM and MIR have so far emphasized stylistic features that relate to formal or structural aspects of lyrics, such as rhyme features, repetition features (e.g., repeated line ratio, echoisms), textual statistics (e.g., word count, POS-tag frequencies) or lexical descriptors (e.g., complexity, readability, type-token ratio, slang frequency) [5, 8, 23, 24]. While such features can capture important stylistic tendencies, they might fail to fully capture genre identity that is also shaped by content-related patterns [6, 9]. In this contribution, we conceptualize content-related genre-discriminating lyrical patterns as recurrent linguistic structures that are strongly associated with specific genres and that relate to the meaning of lyrics. These include topics (What are the lyrics about?, e.g., love, doubt, drugs), sentiments (With what affect do lyrics talk about

© Ananymous Authors. Licensed under a Creative Commons Attribution 4.0 International License (CC BY 4.0). **Attribution:** Ananymous Authors, “"More than words": Genre-Discriminating Lyrical Patterns in Anglophone German Popular Music”, submitted to *ISMIR*, 2026.

something?, e.g., romantic, apocalyptic, playful), and formulaic expressions (genre-specific slang, typical collocations, and vocalizations; e.g., "jah", "get down on it", "oh la la"). Notably, we understand formulaic expressions as a broad category that not only includes multi-word expressions like commonly-used collocations and idioms [25,26], but also single words that show formulaic usage in music lyrics and function as style markers, such as slang words or words with spelling variations [24, 27], highly common words that appear as lexical rather tan topical [24], and single-word vocalizations [28].

Some popular music studies have attempted to capture content-related patterns in large lyrics corpora through topic modeling [29] or sentiment analysis [30]; [31] have investigated the use of formulas in blues lyrics, and [26] have developed an automatic classifier for multi-word idioms in German popular music lyrics. However, to our knowledge so far no study has systematically identified and analyzed genre-discriminating content-related stylistic patterns using machine learning models. To this end, n-gram-based models—despite their conceptual simplicity-directly reflect the recurrent lexical patterns that linguists identify as characteristic of popular song language [32]. At the same time, n-gram-based features have performed comparably to more sophisticated hand-crafted style featrues in lyrics-based genre classification [23, 24].

This study advances lyrics-based MIR by revisiting the n-gram paradigm through an interpretable machine learning lens. We examine a culturally-bounded large corpus of Anglophone popular songs frequently listened to in Germany to identify content-related, genre-discriminative lyrical patterns including topics, sentiments, and formulaic expressions. In doing so, we test whether pattern-based interpretable models can approximate the predictive power of a fine-tuned transformer baseline while providing substantive interpretive value for popular-music research. Specifically, this paper adresses the following research questions:

RQ1 Which content-related genre-discriminative lyrical patterns (topics, sentiments, formulaic expressions) exist in Anglophone German popular music?

RQ2 To what extent do pattern-based interpretable ML models match a fine-tuned transformer baseline on popular music genre prediction?

RQ3 What can we learn about lyrical genre differences from such interpretative models for popular music studies?

2. METHODS

Please refer to Figure 1 for an overview of the method flow. All operations that involved randomness were seeded with 42 for reproducibility; code and data (except for the copyrighted lyrics) will be made publicly available in the camera-ready version.

2.1 Corpus and Genre Annotations

We based our analysis on a previously collected corpus of 254,334 unique popular tracks frequently listened to in Germany over the past 50 years (POPTRAG). The corpus combines multiple sources: (1) 215,702 tracks from weekly *MediaControl* German single and album charts, capturing the 100 most relevant albums and songs per week by physical/digital sales, streams, and broadcasts (18 Nov 1977-21 Nov 2024; limited to 75% of tracks successfully linked to *Spotify* identifiers for consistent metadata); (2) 11,994 unique track entries from Germany's *Daily Top 200 Spotify* charts (Jan 2017–Nov 2024). To capture not only music sales but also listened-to music no longer in the charts, the corpus additionally includes (3) 26,949 *Spotify* track recommendations from 126 German-market genre seeds (retrieved 29 Nov 2024; median 325 per seed, IQR 25, range 49-343); and (4) 8,801 tracks from the 100 top-featured *Spotify* playlists for Germany (retrieved 29 Nov 2024, median 90 tracks per playlist, IQR 42, range 28-750). 853 tracks shorter than 30 seconds were excluded and only unique title-artist combinations were retained, selecting earlier release dates for duplicates.

Using data linkage and webscraping, we enriched POPTRAG with lyrics from *Genius* [1] and release music genre annotations from *Discogs* [2]. We focused our analysis tracks with retrieved English lyrics and at least one *Discogs* music genre annotation (51%). 35% of the remaining tracks were assinged more than one genre by the *Discogs* community. For them, we retained only the most specific genre label (i.e., the one that appeared least frequently in the corpus) to enable an easily interpretable single-label classification task. Finally, we removed genres that made up less than 1% of the remaining corpus, resulting in a final sample of N=126,042 tracks across 9 genres (see Figure 2 for details on the corpus composition).

2.2 RQ1: Music Genre Disrciminating Vocabularies

To identify genre-discriminating content-related lyrical patterns (RQ1), we first performed basic cleaning on raw lyrics from *Genius*: trimming whitespaces, removing content within square brackets, normalizing asterisks, removing repetition and song position indicators, and fixing missing apostrophes (e.g., "don t" → "don't").

To extract relevant n-grams for each lyrical pattern type (topics, sentiments, formulaic expressions), we applied pattern-specific preprocessing steps. For topics and sentiments, we expanded contractions (e.g., "don t" → "do not"), lemmatized and POS-tagged lyrics using *spaCy's* `en_core_web_lg` model with a custom domain lexicon, then reduced to nouns (topics) or adjectives and adverbs (sentiments). We retained only tokens with more than two characters present in both the `en_core_web_lg` vocabulary and the English *nltk* library to exclude non-lexical items. All n-grams were converted to lowercase. We then stratified the corpus into 80% training and 20% holdout sets by genre, ensuring no artist overlap to mitigate artist and album effects [33].

The following steps were applied to the training corpus only; the holdout set was used exclusively for final

[1] https://genius.com
[2] https://discogs.com

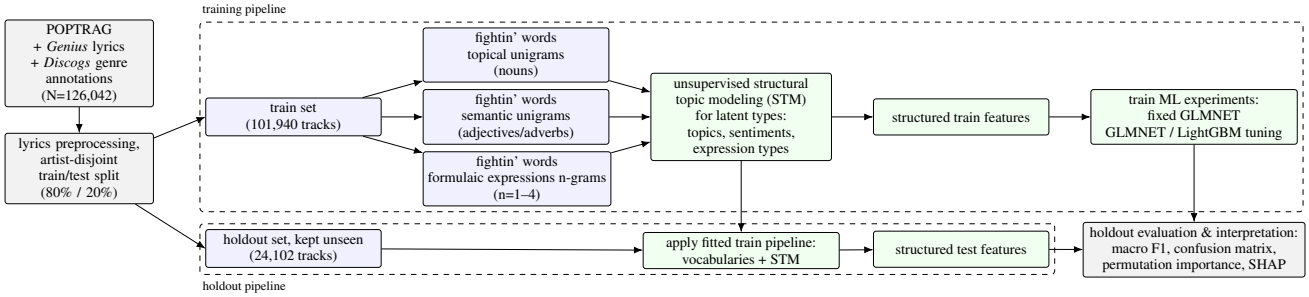


Figure 1: Overview of the methodology. Genre-discriminative vocabularies are constructed on training data only, then integrated via STM into latent interpretable features; fitted components are applied to holdout data for final model evaluation.

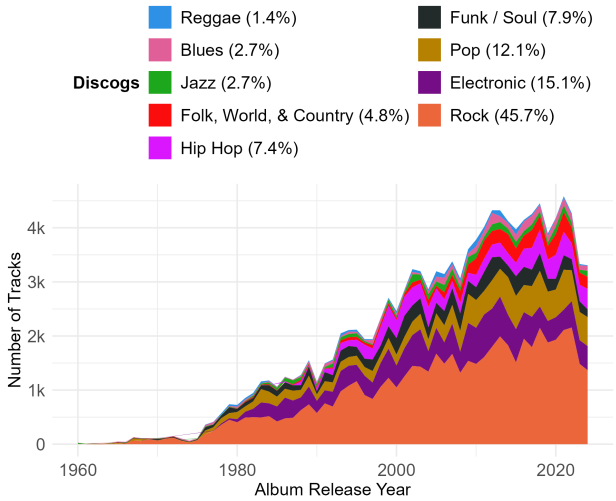


Figure 2: Composition of the analyzed corpus.

ML model evaluation. We constructed three separate vocabularies of genre-discriminating n-grams: unigrams for topics and sentiments, and n-grams (n=1-4) for formulaic expressions. All n-grams required appearance with more than one artist and could not consist solely of stopwords. For expressions, we additionally excluded bigrams starting with "a", "an", "the", and "to" to remove trivial noun+article and to+infinitive constructions.

We then applied the fightin' words method [34] to identify genre-discriminative n-grams. This method quantifies n-gram association with each genre via log-odds ratios—comparing relative frequency in one genre versus all others combined. To account for rare n-grams, we applied Bayesian smoothing using empirical prevalence as a Dirichlet prior. We converted log-odds ratios to z-scores (standardized test statistics indicating standard deviations from zero; larger absolute values indicate stronger discrimination) and applied a one-sided statistical t-test (upper tail) to identify significantly associated n-grams. Given thousands of simultaneous z-score computations across genres, we applied Benjamini-Hochberg false discovery rate (FDR) correction [35] at $\alpha = 0.001$, identifying the most stringent z-score threshold with at most 0.1% expected false positives. N-grams exceeding this FDR-corrected threshold with positive z-scores were retained.

2.3 RQ1: Latent Types of Discriminating Patterns

To identify genre-discriminating latent topics, sentiments, and types of formulaic expressions (RQ1), we employed structural topic modeling (STM) [36] using n-grams from the training corpus vocabularies, guided by target genres as prevalence priors to cluster n-grams into latent types discriminative for specific genres. To mitigate sparsity during model training, we aggregated n-gram counts at the artist level, selecting the most specific genre for artists with multiple genres.

We ran STM separately for topics, sentiments, and expressions, testing an interpretable range of 2 to 20 types each. For determining the number of topics, we used a train-validation split and calculated semantic coherence and likelihood on the validation set as well as the exclusivity score [37] for each configuration. We chose the optimal number of types primarily based on the highest exclusivity score (measuring how uniquely top n-grams associate with single types), while maintaining good semantic coherence and validation likelihood. After selecting the optimal number of types, we reran STM on the full training corpus and interpreted types using n-grams with the highest FREX scores [38], which balance frequency and exclusivity. Finally, we calculated per-genre probabilities for each topic, sentiment, and expression type (mean of type proportions across the genre) in the training set to analyze their distributions.

2.4 RQ2 and RQ3: Machine Learning Experiments

To evaluate whether genre-discriminating lyrical patterns predict music genre (RQ2), we trained machine learning models on extracted topics, sentiments, and expression types as features. We compared performance against a random baseline, a TF-IDF [24] + GLMNET [39] pipeline, and a fine-tuned DistilBERT model [17] trained on raw lyrics. We trained GLMNET models with fixed hyperparameters (penalty $C = 1$, $L1$ ratio$= 0.5$) on individual and combined feature sets. We then performed systematic hyperparameter tuning for combined-features GLMNET (10×10 grid search, C logarithmic $[10^{-5}, 10^{5}]$, $L1$ ratio [0, 1]) and LightGBM [40] (Tree-structured Parzen Estimator [41]: trees [100, 2000], learning rate [0.001, 0.1], leaves [20, 70], max depth [1, 10], minimum leaf data [1, 100], feature fraction [0.1, 1.0], $L1/L2$ regularization loga-

rithmic [10^{-8}, 10.0]) using 4-fold artist-grouped stratified cross-validation. We evaluated performance via macro $F1$ scores from cross-validation and a holdout set untouched by model training or feature extraction.

To identify which patterns actually contribute to genre discrimination and investigate genre profiles (RQ3), we computed permutation feature importance [42] on the holdout set for the best combined-features model. We also computed SHAP values [43] on the holdout set for the best-performing genre to illustrate how specific features drive predictions.

3. RESULTS

3.1 RQ1: Music Genre Discriminating Patterns

Using the fightin' words method on the train corpus, we identified genre-discriminating vocabularies: 3,831 topical nouns, 857 sentiment adjectives and adverbs, and 56,804 formulaic expressions (12% unigrams, 33% bigrams, 35% trigrams, 19% quadgrams). Using STM based on these vocabularies, we identified 10 topics, 6 sentiments, and 11 expression types, which we interpreted and labeled using n-grams with the highest FREX scores (see Table 1 for examples and Appendix for the full list). The genre-discriminative topics were: *Americana*, *Darkness*, *Embodiment*, *Existentialism*, *Heroic Saga*, *Holidays*, *Nature*, *Narrative*, *Romance*, and *Street Life*. Sentiments were: *ambivalent*, *apocalyptic*, *brash*, *nostalgic*, *playful*, and *romantic*. Expression types were: *Christmas Phrases*, *Dance Floor Commands*, *Denials & Negotiations*, *Doom Imaginary*, *Jamaican Patois*, *Landscape Imaginary*, *Love Affirmations*, *Nostalgic Formulas*, *Traditional Street Slang*, *Trap Slang*, and *Vocalizations*.

Per-genre probability distributions of types (see Figure 3) for topics and sentiments show similar distributions with minor variation across genres, though sentiment probabilities are more evenly distributed while topic probabilities concentrate in specific genres. Notably, *Hip Hop* shows distinctively high probability for *Street Life* (41%) and the *brash* sentiment (28%), but comparatively low probabilities for *Existentialism* (13%), *Nature* (4%), *Romance* (6%), and *romantic* sentiment (16%). Similarly, *Rock* exhibits elevated probabilities for *Darkness* (11%), *Heroic Saga* (10%), and *apocalyptic* sentiment (18%). *Existentialism* generally appears consistently (>20%) across nearly all genres. Expression type distributions are more heterogeneous and concentrated than topics and sentiments. Genre-specific concentrations appear for *Traditional Street Slang* (39%) and *Trap Slang* (23%) in *Hip Hop*, *Jamaican Patois* in *Reggae* (33%), *Doom Imaginary* in *Rock* (27%), and *Christmas Phrases* in *Jazz* (21%). In contrast, *Denials & Negotiations*, *Landscape Imaginary*, and *Love Affirmations* distribute more evenly across genres.

3.2 RQ2: Interpretable ML Performance

Using genre-discriminative topics, sentiments, and expression types as lyric features, we trained GLMNET and LightGBM models to predict song genre, comparing performance against a random baseline, TF-IDF + GLMNET, and a fine-tuned DistillBERT model on raw lyrics (Table 2). Our structured representations yielded informative genre signal compared to the random baseline (holdout $F1$=0.111) and uninformed TF-IDF+GLMNET (0.248) pipeline. Among structured features, expressions proved most discriminative (0.282), followed by topics (0.229) and sentiments (0.135). Combined features with GLMNET (C=1, $L1$=0.5) achieved macro $F1$ 0.292 on the holdout, exceeding individual feature sets. Hyperparameter tuning improved performance further. A regularized GLMNET (C =0.002, $L1$=0.33) achieved 0.335 in cross-validation and 0.291 on the holdout set, while the tuned LightGBM performed best among structured approaches (CV 0.354, holdout 0.300). Although DistillBERT remained strongest overall, the tuned LightGBM retained approximately 80% of its predictive power while offering interpretable features and feature contributions.

The tuned LightGBM model performs best for *Hip Hop* (holdout $F1$=0.591) and *Rock* (0.584), and worst for *Folk, World & Country* (0.177) and *Jazz* (0.137) (see Figure 4). Off-diagonal confusions cluster among stylistically adjacent categories (e.g., *Funk / Soul*, *Pop*, *Electronic*, and *Hip Hop*). Although genre prevalence shows a moderate positive relationship with classification quality (Kendall correlation = 0.389), some smaller genres like *Hip Hop* outperform larger ones like *Pop*.

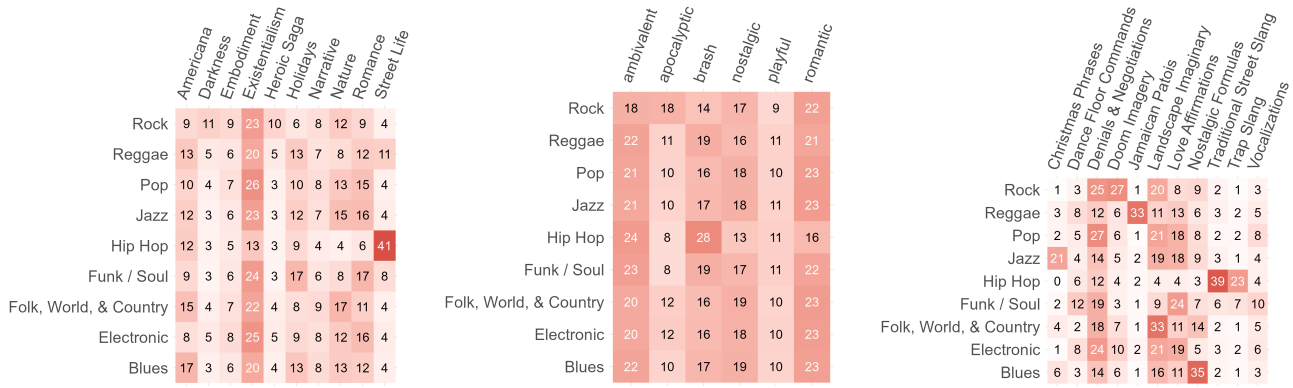
3.3 RQ3: Feature Importances and SHAP Values

To assess which stylistic patterns shape music genre differentiation, we computed permutation feature importance on the holdout set (Figure 5). Expression-type features dominate: *Doom Imagery*, *Traditional Street Slang*, *Love Affirmations*, *Nostalgic Formulas*, *Vocalizations*, and *Dance Floor Commands* are the most influential predictors. Topic features such as *Darkness*, *Narrative*, and *Street Life* occupy intermediate positions, with moderate performance declines when permuted, suggesting thematic context supports but does not solely determine genre distinction. Sentiment features show only minor average impacts. The expression type *Christmas Phrases* displays increased holdout $F1$ when permuted, likely reflecting feature sparsity rather than substantive genre relevance.

As an illustrative example, Figure 6 shows SHAP values for the *Hip Hop* class (holdout $F1$=0.591). High values for *Trap Slang* and *Traditional Street Slang* strongly increase predicted *Hip Hop* scores; *Street Life* and *Denials & Negotiations* similarly shift predictions toward this genre. In contrast, *Christmas Phrases* and *Nostalgic Formulas* are associated with lower *Hip Hop* likelihoods. Some features (e.g., *Doom Imaginary*, *Jamaican Patois*, and *Love Affirmations*) show context-dependent effects, sometimes increasing and sometimes decreasing *Hip Hop* predictions. Sentiment features contribute only minimally to *Hip Hop* identification.

Type	Label	Example high-FREX n-grams
Topic	Americana	beer, john, funk, monkey, whiskey, jack, cowboy, johnny, hat, lee
Topic	Darkness	chaos, death, humanity, destruction, agony, decay, hate, existence, hatred, terror
Sentiment	ambivalent	pretty, maybe, good, nice, bad, sad, easy, glad, totally, busy
Sentiment	apocalyptic	final, evil, eternal, ancient, sacred, deadly, violent, rotten, mortal
Expression	Christmas Phrases	let it snow let, snow let it snow, it snow let it, lonley so lonly so, dreaming of a white, of a white christmas, a merry little christmas, have yourself a merry, yourself a merry little, in a winter wonderland
Expression	Dance Floor Commands	na na na na, doo doo doo doo, ba ba ba ba, ya ya ya ya, man man man man, li li, ma ma ma ma, move your body, put your hands up, ba dum

Table 1: Example interpretations of genre-discriminative topic, sentiments, and expression types with top high-FREX n-grams.



(a) topics (b) sentiments (c) formulaic expressions

Figure 3: Latent genre-discriminative topics, sentiments, and styles: probabilities per genre in the training corpus [%].

Features	Model	Hyperparameters	$F1_{CV}$	$F1_{HO}$
Prevalences	Random	—	—	0.111
Raw Lyrics	DistillBERT	default	—	0.374
TF-IDF	GLMNET	C=1, $L1$=0.5	—	0.248
Topics	GLMNET	C=1, $L1$=0.5	—	0.229
Sentiments	GLMNET	C=1, $L1$=0.5	—	0.135
Expressions	GLMNET	C=1, $L1$=0.5	—	0.282
Combined	GLMNET	C=1, $L1$=0.5	—	0.292
Combined	GLMNET	C=0.002, $L1$=0.33	0.335	0.291
Combined	LightGBM	tuned†	0.354	0.300

Table 2: Macro $F1$ evaluation for classification experiments. Values show mean $F1$ from 4-fold cross-validation (CV) and $F1$ on holdout set (HO). Dashes indicate models trained with fixed hyperparameters. †LightGBM hyperparameters: 223 estimators, learning rate=0.079, 69 leaves, max depth=9, min 31 data points in leaf, L1 Regularization=1.43×10^{-2}, L2 Regularization=9.15×10^{-5}.

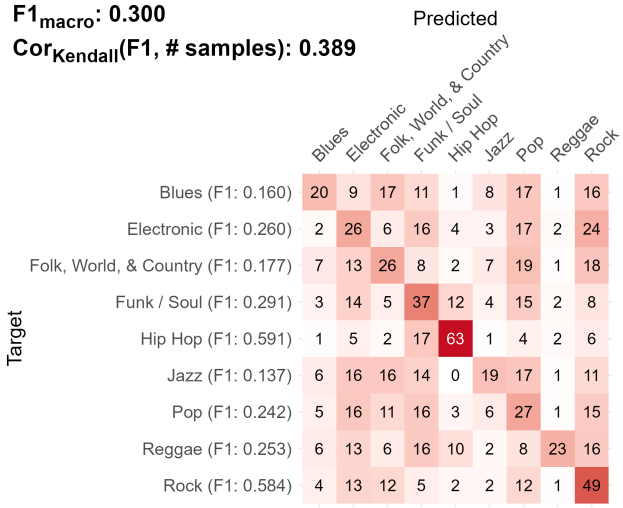


Figure 4: Confusion matrix for combined tuned lightGBM model [%].

4. DISCUSSION

In this contribution, we identify genre-discriminative and content-related lyrical patterns in Anglophone popular music frequently listened to in Germany (RQ1): 10 topics, 6 sentiments, and 11 types of formulaic expressions. Interpretable ML models based on these patterns (RQ2) retain 80% of the predictive performance of a fine-tuned DistilBERT classifier, demonstrating that pattern-based features effective genre classification with interpretability. Feature importance analyses (RQ3) reveal that formulaic expressions are substantially more discriminative than topics and sentiments. Using *Hip Hop* as an example, we show how expressions, topics, and sentiments jointly characterize this genre.

Lyrical genre distinction is driven not only by lexical, diversity, readability, rhyme, structural, and emotional descriptors [5, 8, 23, 24], but also by topical and formulalic

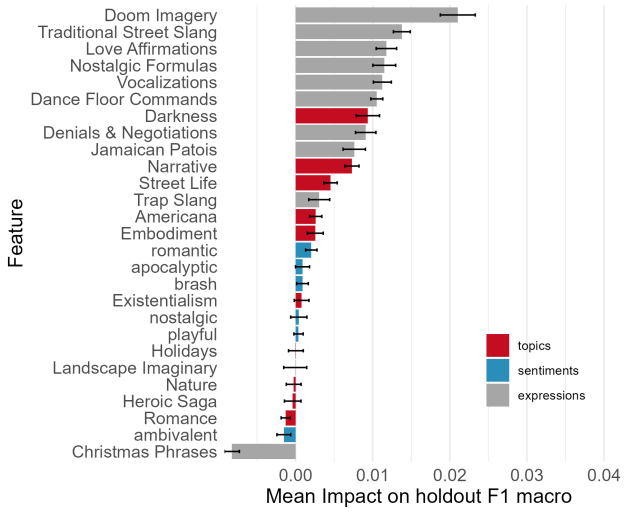


Figure 5: Permutation importance on Heldout data for the tuned combined features LightGBM model.

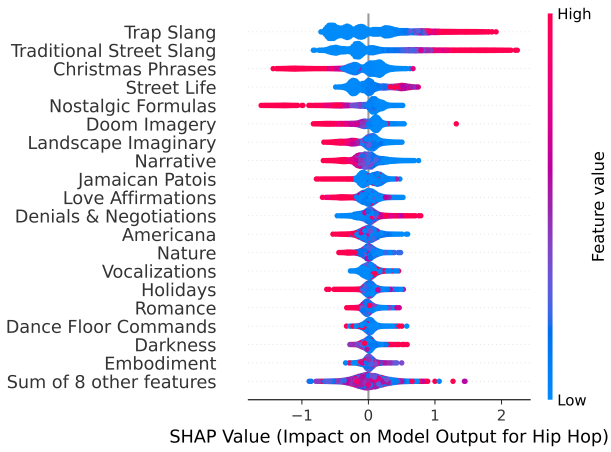


Figure 6: Example: SHAP values for *Discogs' Hip Hop*.

patterns identified in our analysis (RQ1). Formulaic expression types emerge as particularly informative for genre differentiation compared to topics and sentiments, supporting the view that genre-specific phrasing contributes substantially to how genres are lyrically defined and perceived [32]

In our interpretable ML experiments (RQ2), Hip Hop and Rock were most readily identifiable from lyrical features alone, suggesting some genres rely more strongly on distinctive lyrical markers while others share markers and are more easily confused. Given that *Discogs' Rock* encompasses metal styles, this aligns with prior lyric-based genre work [10, 24] and the view of genre as a fuzzy category organized around prototypical features [4]. Importantly, our human-readable features recovered substantial DistilBERT performance, indicating the transformer exploits structure accessible to interpretable n-gram analysis.

Feature importance analysis (RQ3) reveals genre differences are primarily driven by formulaic expression types like *Traditional Street Slang*, *Trap Slang*, and *Doom Imagery* rather than emotion or theme alone. The *Hip Hop* case illustrates this: street life topics and slang expressions jointly elevate predictions. Genres differ in how they package meaning through stylistic phrasing, with thematic content playing a supporting role. Interactions between expression types, topics, and sentiments form lyrical profiles that encode genre identity—consistent with viewing genre as a fuzzy, relational construct distinguished by bundles of linguistic cues and shaped by shared societal conventions [4].

Beyond insights into Anglophone German popular music, our findings advance CM and MIR by establishing a principled division between predictive and explanatory ML approaches. While black-box transformers excel at predictive performance, interpretable models reveal which linguistic patterns drive predictions, enabling insights into lyrical genre formation rather than mere tagging [18]. This explanatory capacity supports practical applications: understanding genre biases in generative lyrics models by identifying strong signals like Trap Slang and Doom Imagery to mitigate stylistic skews [22]; corpus curation tools that flag culturally distinctive markers for ethnomusicological analysis [31]; and hybrid genre taggers for streaming platforms balancing predictive power with transparency.

Our analysis faces several limitations. First, music genre is inherently fuzzy [44]; by using *Discogs* labels, we adopt a particular community's perspective on genre boundaries. We intentionally simplified the multi-label nature of tracks into single-label classification to enable interpretable modeling. Second, we adopted an aggregate view without accounting for temporal trends within genres' lyrics. Third, permutation feature importance can be biased by correlated features [45]. Although some features showed correlation, we preserved individual linguistic pattern interpretability at the cost of potential ranking distortion. Finally, our corpus focuses on Anglophone popular music consumed in Germany, limiting generalizability to global Anglophone repertoires or non-English music popular in Germany.

Future research will extend the presented approach to other languages, comparing lyrical patterns across linguistic contexts and exploring language-agnostic interpretable models. We also plan to test generalizability across diverse corpora beyond German popular music listening and incorporate additional lyrical features alongside multimodal audio characteristics [12] to build more robust interpretable genre classifiers.

5. AI USAGE STATEMENT

See submission guidelines. Does not count towards the page limit.

6. REFERENCES

[1] S. Davis, "Pop lyrics: A mirror and a molder of society," *Etc-review of General Semantics*, vol. 42, p. 167, 1985.

[2] S. Frith, "Why Do Songs Have Words?" *The Socio-*

logical Review, vol. 34, no. 1_suppl, pp. 77–106, May 1986.

[3] K. Watanabe and M. Goto, "Lyrics Information Processing: Analysis, Generation, and Applications," in *Proceedings of the 1st Workshop on NLP for Mu- Sic and Audio (NLP4MusA)*. Online: Association for Computational Linguistics, Oct. 2020, pp. 6–12.

[4] F. Fabbri, "A Theory of Musical Genres: Two Applications," in *Popular Music Perspectives*, D. Horn and P. Tagg, Eds. Göteborg & Exeter: IASPM, 1982, pp. 52–81.

[5] E. Parada-Cabaleiro, M. Mayerl, S. Brandl, M. Skowron, M. Schedl, E. Lex, and E. Zangerle, "Song lyrics have become simpler and more repetitive over the last five decades," *Scientific Reports*, vol. 14, no. 1, p. 5531, Mar. 2024.

[6] L. Carbone, P. Alvarez-Cueva, and L. Vandenbosch, "Status Markers in Popular Music Across Six Countries: A Content Analysis of Gender, Race/Ethnicity, Genre, and Capital in Music Lyrics," *Sex Roles*, vol. 90, no. 7, pp. 891–909, Jul. 2024.

[7] L. Betti, C. Abrate, and A. Kaltenbrunner, "Large scale analysis of gender bias and sexism in song lyrics," *EPJ Data Science*, vol. 12, no. 1, pp. 1–22, Dec. 2023.

[8] M. Schedl, "Genre Differences of Song Lyrics and Artist Wikis: An Analysis of Popularity, Length, Repetitiveness, and Readability," in *The World Wide Web Conference*, ser. WWW '19. New York, NY, USA: Association for Computing Machinery, May 2019, pp. 3201–3207.

[9] L. Squires, "Genre and linguistic expectation shift: Evidence from pop song lyrics," *Language in Society*, vol. 48, no. 1, pp. 1–30, Feb. 2019.

[10] A. Tsaptsinos, "Lyrics-Based Music Genre Classification Using a Hierarchical Attention Network," in *18th International Society for Music Information Retrieval Conference*, Suzhou, 2017.

[11] R. Mayer, R. Neumayer, and A. Rauber, "Rhyme and Style Features for Musical Genre Classification by Song Lyrics," in *Proceedings of the 9th International Society for Music Information Retrieval Conference*, Philadelphia, 2008, pp. 337–342.

[12] Y. Li, Z. Zhang, H. Ding, and L. Chang, "Music genre classification based on fusing audio and lyric information," *Multimedia Tools and Applications*, vol. 82, no. 13, pp. 20 157–20 176, May 2023.

[13] H. Akalp, E. Furkan Cigdem, S. Yilmaz, N. Bolucu, and B. Can, "Language Representation Models for Music Genre Classification Using Lyrics," in *2021 International Symposium on Electrical, Electronics and Information Engineering*, ser. ISEEIE 2021. New York, NY, USA: Association for Computing Machinery, Jul. 2021, pp. 408–414.

[14] R. de Araújo Lima, R. C. C. de Sousa, H. Lopes, and S. D. J. Barbosa, "Brazilian Lyrics-Based Music Genre Classification Using a BLSTM Network," in *Artificial Intelligence and Soft Computing*, L. Rutkowski, R. Scherer, M. Korytkowski, W. Pedrycz, R. Tadeusiewicz, and J. M. Zurada, Eds. Cham: Springer International Publishing, 2020, pp. 525–534.

[15] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of Deep Bidirectional Transformers for Language Understanding," in *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, J. Burstein, C. Doran, and T. Solorio, Eds. Minneapolis, Minnesota: Association for Computational Linguistics, Jun. 2019, pp. 4171–4186.

[16] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A Robustly Optimized BERT Pretraining Approach," Jul. 2019.

[17] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," in *5th Workshop on Energy Efficient Machine Learning and Cognitive Computing - NeurIPS 2019*. arXiv, Mar. 2020.

[18] N. M. Gardazi, A. Daud, M. K. Malik, A. Bukhari, T. Alsahfi, and B. Alshemaimri, "BERT applications in natural language processing: A review," *Artificial Intelligence Review*, vol. 58, no. 6, p. 166, Mar. 2025.

[19] B. Škrlj, S. Sheehan, N. Eržen, M. Robnik-Šikonja, S. Luz, and S. Pollak, "Exploring Neural Language Models via Analysis of Local and Global Self-Attention Spaces," in *Proceedings of the EACL Hackashop on News Media Content Analysis and Automated Report Generation*, H. Toivonen and M. Boggia, Eds. Online: Association for Computational Linguistics, Apr. 2021, pp. 76–83.

[20] E. M. Bender, T. Gebru, A. McMillan-Major, and S. Shmitchell, "On the Dangers of Stochastic Parrots: Can Language Models Be Too Big?" in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency*, ser. FAccT '21. New York, NY, USA: Association for Computing Machinery, Mar. 2021, pp. 610–623.

[21] B. L. Sturm, "A Simple Method to Determine if a Music Information Retrieval System is a "Horse"," *IEEE Transactions on Multimedia*, vol. 16, no. 6, pp. 1636–1644, Oct. 2014.

[22] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nature Machine Intelligence*, vol. 1, no. 5, pp. 206–215, May 2019.

[23] M. Mayerl, M. Vötter, M. Moosleitner, and E. Zangerle, "Comparing Lyrics Features for Genre Recognition," in *Proceedings of the 1st Workshop on NLP for Music and Audio (NLP4MusA)*, S. Oramas, L. Espinosa-Anke, E. Epure, R. Jones, M. Sordo, M. Quadrana, and K. Watanabe, Eds. Online: Association for Computational Linguistics, 2020, pp. 73–77.

[24] M. Fell and C. Sporleder, "Lyrics-based analysis and classification of music," in *Proceedings of COLING 2014, the 25th International Conference on Computational Linguistics: Technical Papers*, 2014, pp. 620–631.

[25] A. Wray and M. R. Perkins, "The functions of formulaic language: An integrated model," *Language & Communication*, vol. 20, no. 1, pp. 1–28, Jan. 2000.

[26] M. Amin, P. Fankhauser, M. Kupietz, and R. Schneider, "Data-driven Identification of Idioms in Song Lyrics," in *Proceedings of the 17th Workshop on Multiword Expressions (MWE 2021)*, P. Cook, J. Mitrović, C. P. Escartín, A. Vaidya, P. Osenova, S. Taslimipoor, and C. Ramisch, Eds. Online: Association for Computational Linguistics, Aug. 2021, pp. 13–22.

[27] R. Kreyer and J. Mukherjee, "The Style of Pop Song Lyrics: A Corpus-linguistic Pilot Study," *Anglia*, vol. 125, no. 1, pp. 31–58, Aug. 2007.

[28] C. Knauf, ""La La La, I'm not listening (but I really am)": Exploring the meaning of a rhythmic hook in rock and pop music," *Atlantic Journal of Communication*, vol. 33, no. 3, pp. 525–545, May 2025.

[29] A. Bansal, R. Agarwal, and K. Jain, "Linguistic Complexity and Socio-cultural Patterns in Hip-Hop Lyrics," Apr. 2025.

[30] K. Napier and L. Shamir, "Quantitative Sentiment Analysis of Lyrics in Popular Music," *Journal of Popular Music Studies*, vol. 30, no. 4, pp. 161–176, Dec. 2018.

[31] N. Stoia, "Blues Lyric Formulas in Early Country Music, Rhythm and Blues, and Rock and Roll," *Music Theory Online*, vol. 26, no. 4, Nov. 2020.

[32] V. Werner, "Catchy and conversational? A register analysis of pop lyrics," *Corpora*, vol. 16, no. 2, pp. 237–270, Aug. 2021.

[33] A. Flexer and D. Schnitzer, "Effects of Album and Artist Filters in Audio Similarity Computed for Very Large Music Databases," *Computer Music Journal*, vol. 34, no. 3, pp. 20–28, 2010.

[34] B. L. Monroe, M. P. Colaresi, and K. M. Quinn, "Fightin' Words: Lexical Feature Selection and Evaluation for Identifying the Content of Political Conflict," *Political Analysis*, vol. 16, no. 4, pp. 372–403, 2008.

[35] Y. Benjamini and Y. Hochberg, "Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.

[36] M. E. Roberts, B. M. Stewart, and D. Tingley, "Stm: An R Package for Structural Topic Models," *Journal of Statistical Software*, vol. 91, pp. 1–40, Oct. 2019.

[37] M. E. Roberts, B. M. Stewart, D. Tingley, C. Lucas, J. Leder-Luis, S. K. Gadarian, B. Albertson, and D. G. Rand, "Structural Topic Models for Open-Ended Survey Responses," *American Journal of Political Science*, vol. 58, no. 4, pp. 1064–1082, 2014.

[38] J. M. Bischof and E. M. Airoldi, "Summarizing topical content with word frequency and exclusivity," in *Proceedings of the 29th International Coference on International Conference on Machine Learning*, ser. ICML'12. Madison, WI, USA: Omnipress, Jun. 2012, pp. 9–16.

[39] A. Defazio, F. Bach, and S. Lacoste-Julien, "SAGA: A Fast Incremental Gradient Method With Support for Non-Strongly Convex Composite Objectives," Dec. 2014.

[40] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems*, vol. 30. Curran Associates, Inc., 2017.

[41] T. Akiba, S. Sano, T. Yanase, T. Ohta, and M. Koyama, "Optuna: A next-generation hyperparameter optimization framework," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2019.

[42] A. Fisher, C. Rudin, and F. Dominici, "All Models are Wrong, but Many are Useful: Learning a Variable's Importance by Studying an Entire Class of Prediction Models Simultaneously," *Journal of machine learning research : JMLR*, vol. 20, p. 177, 2019.

[43] S. M. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," in *Advances in Neural Information Processing Systems*, vol. 30. Curran Associates, Inc., 2017.

[44] A. van Venrooij and V. Schmutz, "Categorical ambiguity in cultural fields: The effects of genre fuzziness in popular music," *Poetics*, vol. 66, pp. 1–18, 2018.

[45] C. Molnar, *Interpretable Machine Learning*. Lulu.com, 2020.